Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Senior AI engineer with **9+ years delivering production software** in regulated enterprise environments, now specializing in agentic AI and intelligent automation. At Intuit, designs and deploys **production-grade agentic AI solutions** using Anthropic Claude, Python/FastAPI, RAG pipelines, and cloud-native services - owning the full lifecycle from architecture through deployment and operational support. Built guardrails for **prompt injection defense, PII filtering, and AI security controls** in regulated financial data environments. At Entrata (100K+ MAU platform), owned **PHI/HIPAA-compliant data infrastructure** on AWS, Terraform IaC, and CI/CD - shipping and operating production services in a highly regulated environment. Built **Leasel**: an 8-agent LangGraph system with tool use, RAG (document ingestion, vector search, semantic retrieval), and LLM-as-judge evaluation (93.6% precision). Strong in **Python, FastAPI, Docker, Databricks, Azure AI Foundry, GCP Vertex AI, Anthropic Claude, LLMOps**. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Agentic AI & Intelligent Automation | Anthropic Claude / Azure AI Foundry / GCP Vertex AI | RAG Pipelines (Ingestion, Vector Search, Reranking) | Tool Use / Function Calling / Agent Orchestration | Prompt Injection Defense & AI Security Controls | Python / FastAPI / SQL / REST APIs | Databricks / Snowflake / ETL Pipelines | LLMOps / CI/CD Eval Pipelines / Golden Datasets | PHI / HIPAA / Regulated Enterprise Environments | AWS / Docker / Kubernetes / Terraform / GitHub Actions | End-to-End Solution Lifecycle Ownership | Cross-functional Collaboration & Technical Mentorship

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - Agentic AI Solutions - Virginia, USA

- **Design, develop, and deploy production-grade agentic AI solutions** using Anthropic Claude, Python/FastAPI, LangGraph orchestration, and Azure AI Foundry - owning the full solution lifecycle from prototype through production deployment and continuous optimization.
- Build **RAG components**: document ingestion, embedding generation, vector indexing (ChromaDB, Pinecone), retrieval optimization, reranking, and semantic search for enterprise financial workflows.
- Implement **secure-by-design AI architectures**: prompt injection safeguards, PII filtering/redaction, AI security controls, resiliency patterns, and graceful degradation strategies for regulated financial data environments.
- Build **LLMOps evaluation pipelines** (LLM-as-judge + golden datasets) integrated into GitHub Actions CI/CD; make pragmatic architectural decisions on AI-driven vs. traditional automation approaches based on production requirements.
- Collaborate with Architecture, Platform Engineering, Product, and Operations teams; provide technical mentorship and drive adoption of agentic AI patterns across the organization.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users · Python, Java, AWS, Terraform

- Owned **PHI/regulated data infrastructure** in a HIPAA-compliant environment: AWS S3 migration with Libsodium encryption, access management, and audit logging - the same regulated enterprise environment CVS Health requires.
- Delivered **intelligent automation solutions** end-to-end: Python/Bash automation eliminating 80% of manual operational work; internal debugging tools reducing triage time by 40%; large-scale data pipelines processing 1.6M+ records.
- Owned **Terraform IaC and GitHub Actions CI/CD**: Docker microservices on AWS (EC2, RDS, S3, Lambda, SQS, IAM) with full environment promotion and operational reliability practices.

- Optimized 200+ SQL queries (30% gain); built ETL pipelines with validation and rollback checks; reviewed 300+ PRs and mentored junior engineers across 120+ Agile sprints.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - **Pune, India**

- Built production Python/Java/PostgreSQL modules for regulated housing workflows; delivered large-scale ETL pipelines (1.6M records) with validation and reconciliation.

PROJECTS

Leasel - 8-Agent Agentic AI System

Python · LangGraph · FastAPI · RAG · Anthropic Claude · LLMOps · HIPAA-style Controls

Production-grade agentic AI system: **tool use, function calling, agent orchestration** (LangGraph), document ingestion (OCR), vector search with reranking, structured output validation, and **AI security controls** routing high-stakes decisions away from the LLM. LLMOps evaluation harness (LLM-as-judge + golden datasets) integrated into CI/CD: 93.6% precision, 87.9% F1 across 1,240 test cases. Demonstrates full solution lifecycle ownership from architecture to production in a regulated domain.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - LLMOps Evaluation Platform

Python · FastAPI · Anthropic Claude · 5 LLM Providers · CI/CD · Golden Datasets

15-endpoint FastAPI service integrating **Anthropic Claude** and four other LLM providers. Evaluates agentic AI outputs on factual accuracy, hallucination rate, retrieval quality, and compliance. **LLMOps CI/CD pipeline** with automated regression tests and golden datasets - the same evaluation infrastructure needed for enterprise AI governance and responsible AI practices.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017